GateTunnel

System Design Description

Eclipse Arrowhead 5.2.2 | support system | Revision 2026-11-09 | REVIEW DRAFT

Purpose and document structure

This document describes implementation ownership, architecture, persistence, security and operational boundaries. The current cross-language sections below precede the detailed implementation appendices. Appendix dates and release labels belong to the retained baseline, not to a newly qualified release. Existing descriptions have been assembled from the primary trees; no live interoperability or semantic-parity claim is made.

Java implementation

Primary folder: Eclipse-Arrowhead-java/ah5-GateTunnel-java-spring. 1 detailed Java SysDD source document(s) are included in the appendices. Their architecture, operation parameters, models, transport bindings, access rules and errors remain associated with Java. They must not be read as automatically implemented by the ports.

Go implementation

Primary folder: Eclipse-Arrowhead-go/ah5-GateTunnel-go.

GateTunnel - Go 5.2.2

Standalone support system, reorganized on 2026-10-28 following the Eclipse-Arrowhead-java distribution layout. Build and operate from this folder; no original Arrowhead checkout or sibling source module is required. The executable is bin/GateTunnel. Entry point: ./cmd/gatetunnel. Module: arrowhead/ah5-support-go/gatetunnel. See DEPLOYMENT.md for configuration and standalone container use. Existing compatibility limits remain in the copied upstream documentation and UPSTREAM-README.md. The dedicated launcher selects only this support-system role. The formerly shared internal runtime and regression tests are copied locally, not linked from another tree. Files for common helpers and other route implementations remain in that private runtime; this packaging does not claim new functionality or Java parity. Documentation and license notices are included in this folder. Source provenance and import-only rewrites are listed in the distribution's SOURCE-MANIFEST.json. No running cloud or upstream source was changed.

Listening port

Built-in default 127.0.0.1:8453 (Arrowhead default). The shipped config.json, compose.yaml and Dockerfile use the same port on all interfaces ("listen": "0.0.0.0:8453"; 8540-8555 layout before 2026-11-09). Override with "listen" in the configuration file, SERVER_PORT=N or -server.port N; the port overrides keep the configured host. See [LISTENING-PORTS.md](../LISTENING-PORTS.md) for precedence and all systems.

Documented behavior and remaining work

System	Implemented behavior	Remaining work
GateTunnel	Legacy-mTLS relay forwarding, owner-constrained session control, expired-session rejection	AH-GT-1 data plane; control-session rows are not additional AMQP tunnels; listeners are volatile and renegotiated on restart

Detailed implementation evidence

1 native Go SysDD document(s) are included in the appendices. Their protocol and storage descriptions apply to that implementation only. Current primary-folder ownership and current listening-port notes above supersede historical paths or example ports in those retained documents.

Rust implementation

Primary folder: Eclipse-Arrowhead-rust/ah5-GateTunnel-rust.

GateTunnel (Rust)

Arrowhead 5.2.2 GateTunnel support system. Entry point src/main.rs is built as binary gatetunnel by the shared crate in ../ah5-support-runtime-rust (arrowhead-support-rust).

- examples/gatetunnel.json - example configuration (master copy: ../ah5-support-runtime-rust/examples/)

- documentation/ - SysD/SysDD/IDD/SD sources (symlink into ../ah5-support-runtime-rust/documentation/GateTunnel, where the contract generator writes)

Run: cargo run -p arrowhead-support-rust --bin gatetunnel -- <config.json>

Listening port

Default 127.0.0.1:8453 (Arrowhead default, role gatetunnel). Override with "listen": "host:port" in the configuration, SERVER_PORT=N or --server.port N; the port overrides keep the configured host. See [LISTENING-PORTS.md](../LISTENING-PORTS.md) for precedence and all systems.

Documented behavior and remaining work

System	Implemented behavior	Remaining work
GateTunnel	Signed AH-GT-1 exchange, dynamic mTLS listeners, encrypted forwarding, bounded sessions, replay and scoped operation guards	Exhaustive DTO/chaos tests, independent review; segment-exact grants are stricter than reference descendant grants

Detailed implementation evidence

1 native Rust SysDD document(s) are included in the appendices. Their protocol and storage descriptions apply to that implementation only. Current primary-folder ownership and current listening-port notes above supersede historical paths or example ports in those retained documents.

Build and state ownership

Core entry points are built by arrowhead-core-rust; the ordinary support entry points are built by arrowhead-support-rust using the shared support runtime. LocalCloudManager has its own crate. Per-system documentation links may point into the shared documentation catalog. Independent identity and state configuration is still required. Read the native appendix for persistence and restart semantics.

Cross-language qualification

Identical system names or HTTP paths do not establish interoperability. For the selected implementation, verify registered service versions and metadata, method/path and payload models, null and optional-field handling, success/error bodies, authentication and per-operation authorization, timeout/retry behavior and persistence across restart. Java MQTT/AMQP descriptions apply to a port only where its own implementation evidence explicitly supports them. Missing transports must not be advertised. State conversion from Java SQL to native snapshots, secret provisioning, backup/restore and rollback require implementation-specific procedures. No data migration, deployment or cloud reconfiguration was performed for this document set.

Detailed appendices and provenance

- Java: Eclipse-Arrowhead-java/ah5-GateTunnel-java-spring/documentation/v5.2.2/GateTunnel-v5.2.2-SysDD.tex

- Go: Eclipse-Arrowhead-go/ah5-GateTunnel-go/documentation/GateTunnel/GateTunnel-SysDD.tex

- Rust: Eclipse-Arrowhead-rust/ah5-support-runtime-rust/documentation/GateTunnel/GateTunnel-SysDD.tex

The package sources/ folder retains the selected LaTeX, figures, bibliography, language README/build evidence and source snapshots for newly described contracts. source-manifest.json records SHA-256 hashes of original inputs. build-report.json records whether an appendix used an existing newer PDF or was compiled from source.

GateTunnel

System Design Description

Abstract
This document describes the design and implementation of the GateTunnel support system of Arrowhead 5.2.2 (`ah5-GateTunnel-java-spring`).

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1 Overview

This document describes the GateTunnel system implementation, which realizes the GateTunnel SysD [1]. GateTunnel carries the inter-cloud service calls negotiated by the Gatekeepers: a consumer session listens locally and forwards each request over the relay to the provider cloud's GateTunnel, which calls the provider. In Section 2, we describe implementation details of the system.

2 Implementation

This implementation realizes the GateTunnel SysD in `ah5-GateTunnel-java-spring`. It follows the Arrowhead Java-Spring system pattern of the support systems (as Blacklist and EventHandler). The Java sources, the private copy of the Arrowhead common library (`vendor/`), tests, images and deployment definition are owned by this folder. The earlier native implementation (JDK `HttpsServer`, SQLite, package `eu.arrowhead.port`) has been replaced; the wire contracts are unchanged. `legacy/` preserves an older implementation and is excluded from the build.

2.1 Implementation language and tools

- Java 21, Spring Boot 3.4 (web, security, data JPA, Quartz), the vendored Arrowhead common library (common-utils, data-transfer-objects), Bouncy Castle, Eclipse Paho MQTT v3 (relay client) and the RabbitMQ AMQP client (sysMonitor). Maven builds the Spring Boot JAR `arrowhead-gatetunnel-5.2.2.jar`; the POM inherits only the Spring Boot parent.
- Storage: a private MySQL 8.4 schema `ah_gatetunnel` through JPA (MariaDB driver, `sslMode=verify-full`, `ddl-auto=validate`).

2.2 Functional properties implementation

The package `eu.arrowhead.gatetunnel` holds the following components:

Component	Responsibility
`GateTunnelSystemInfo`, `init/`	Registered system and services; at start, the system certificate must be `GateTunnel.<cloud>.<organisation>.arrowhead.eu` with profile `sy`. Registration itself is done by the common `ApplicationInitListener`.
`api/http/`	Controllers of gateTunnelConnection, gateTunnelKey, gateTunnelManagement, generalManagement, echo and liveness. Each path is served under `/gatetunnel`, `/gateway` and without base path.
`security/GateTunnelCallerFilter`	Caller rules after the common authentication filter: certificate profile `sy`/`op`; management paths for LocalCloudManager and Sysop; other operations for Gatekeeper, DynamicServiceOrchestration, SimpleStoreServiceOrchestration, LocalCloudManager and Sysop; inter-cloud sessions for the local Gatekeeper only.
`service/InterCloudSessionService`	Inter-cloud relay sessions (in memory): provider side executes relayed requests within the negotiated operations; consumer side runs a mutual-TLS tunnel listener per session in the pool 8200–8299 and forwards over the relay. Ported verbatim in behaviour from 5.2.2.
`service/SessionSecurity`, `PeerNames`, `validation/OperationScope`	AH-GT-1 key exchange, sealing and replay window; explicit cross-organisation bindings; scope of a relayed request. Byte-for-byte compatible with the Go and Rust GateTunnels.
`service/RelayClient`	Paho MQTT client with the GateTunnel's own relay identity (`RELAY_*`); topics `arrowhead/relay/<cloud>/gatetunnel/in/<sender CN>/<id>`.
`service/ProviderClient`	Calls the negotiated provider with the own identity token, pinned to the provider's certificate CN, renewing a refused token once.

Component	Responsibility
`service/ControlSession Service, jpa/`	4.x-compatible control sessions (control port pool 8000–8099).
`service/GateTunnelJobs`	Scheduled relay (re)connection (10 s) and session expiry (60 s).

2.2.1 Data handled and stored

Storage: MySQL schema `ah_gatetunnel`:

Tables	Content
`session`	Control sessions: session and queue identifiers, peer name, service definition, reserved port, state, parties as JSON. Closed sessions are removed after `SESSION_MAX_AGE_SECONDS`, which frees their ports.
`logs`	Log entries (Log4j2 JDBC appender), read by general-Management `get-log`.

Inter-cloud sessions and their keys are kept in memory only. The interfaces exchange JSON objects; their structure is specified in the IDD documents. Error responses use the Arrowhead error body with the 5.2.2 status codes.

2.3 Non functional properties implementation

2.3.1 Security

The web server requires client certificates (PKCS12 key and trust stores) and checks them against the cloud CA's CRL. The common `OutsourcedFilter` verifies the identity token with Authentication and binds it to the client certificate; the caller filter then applies the rules above. Tunnel listeners and provider calls use the same identity and CRL outside the web server. `SecurityProfileGuard` refuses a configuration below the secure profile. The database account has data-manipulation privileges only.

2.3.2 Power management

None; the system is a server process without power states. The `sysMonitor power` observation always reports line supply.

2.3.3 Internal monitoring

Logs go to stdout, a log file and the `logs` table. The common `SysMonitorPublisher` publishes the `sysMonitor` observations; the health contribution is `degraded` while the relay link is down.

2.3.4 Configuration

`config/application.properties`; the 5.2.2 environment names (`LOCAL_CLOUD`, `ADVERTISED_ADDRESS`, `NEIGHBOR_CLOUDS`, `FEDERATION_*`, `RELAY_*`, port pools, session times) are placeholders in it. Secrets are Compose secrets read through `configtree:/run/secrets/`.

2.4 Deployment

The local `Dockerfile` builds this folder and runs `arrowhead-gatetunnel-standalone:5.2.2` as UID 10001 with a read-only root filesystem. `compose.yaml` adds `gatetunnel-db` (MySQL 8.4.11, TLS required, internal network; `deployment/database/init.sql` and `least-privilege.sh`) and joins an operator-selected cloud network. The API uses port 8453; the health check is a TCP probe of it (the former plain-HTTP

health port 8481 is gone). Certificates and credentials are mounted individually. Existing test-cloud deployments are not changed by this packaging.

ServiceRegistry is the bootstrap endpoint; Authentication and the orchestrator are found through it. Federation requires the organisation MQTT relay, configured neighbours, the relay identity and organisation trust. The Compose default is AH-GT-1 with explicit legacy-mTLS compatibility; peers must agree, and there is never a silent downgrade.

2.4.1 Ownership and build boundary

The Arrowhead common library is a private vendored source copy (`vendor/PROVENANCE.json`); security fixes must be propagated to every vendored copy. There is no shared runtime process, database or Maven parent besides Spring Boot's. Build with Java 21 and `mvn -B clean verify` (`scripts/verify.sh` also checks the JAR contents), or with the local Docker build.

2.5 Local modifications

Notes on this implementation in the workspace:

- Java-Spring port of the native implementation; inter-cloud sessions for DynamicServiceOrchestration (local extension).
- Services registered with version `5.2.2`; generalManagement adds `get-log` and `get-config`.
- `/health` now requires an identity token; unprefixed path aliases are served.

2.6 Verification

JUnit tests: the AH-GT-1 reference vector (key exchange, key derivation, sealing, replay), peer names, operation scope, TLS pinning and token renewal, the refused insecure start, and a MockMvc contract test of every served path and method through the real authentication filter. Integration tests against containers: the application against a fresh database container (schema validated with DML-only grants), and an AH-GT-1 round trip between two relay identities over an MQTT broker with mutual TLS. A live federation run with the Go and Rust GateTunnels has not been repeated for the Spring build.

3 Services

The implementation of the services is based on the SD and IDD documents referenced below.

Services produced	**SysD ref**	**SD ref**	**IDD ref**
`gateTunnelConnection`	[1]	[2]	[3]
`gateTunnelKey`	[1]	[4]	[5]
`gateTunnelManagement`	[1]	[6]	[7]
`sysMonitor`	[1]	[8]	[9]
Services consumed	**SysD ref**	**SD ref**	**IDD ref**
serviceDiscovery	[10]	[11]	–
identity	[12]	[13]	–
inter-cloud messages	Relay (MQTT)	–	–
negotiated service	Providers	–	–
sysMonitor (publisher)	AMQP broker (SysMonitor)	–	–

4 References

[1] J. Delsing, “GateTunnel – System Description,” Eclipse Arrowhead, SysD 5.2.2, 2026, ah5-GateTunnel-java-spring/documentation/v5.2.2/GateTunnel-v5.2.2-SysD.pdf.

[2] ——, “gateTunnelConnection – Service Description (GateTunnel),” Eclipse Arrowhead, SD 5.2.2, 2026, gateTunnel-v5.2.2-gateTunnelConnection-SD.pdf.

[3] ——, “gateTunnelConnection HTTP/TLS/JSON – Interface Design Description (GateTunnel),” Eclipse Arrowhead, IDD 5.2.2, 2026, gateTunnel-v5.2.2-gateTunnelConnection-HTTP-IDD.pdf.

[4] ——, “gateTunnelKey – Service Description (GateTunnel),” Eclipse Arrowhead, SD 5.2.2, 2026, gateTunnel-v5.2.2-gateTunnelKey-SD.pdf.

[5] ——, “gateTunnelKey HTTP/TLS/JSON – Interface Design Description (GateTunnel),” Eclipse Arrowhead, IDD 5.2.2, 2026, gateTunnel-v5.2.2-gateTunnelKey-HTTP-IDD.pdf.

[6] ——, “gateTunnelManagement – Service Description (GateTunnel),” Eclipse Arrowhead, SD 5.2.2, 2026, gateTunnel-v5.2.2-gateTunnelManagement-SD.pdf.

[7] ——, “gateTunnelManagement HTTP/TLS/JSON – Interface Design Description (GateTunnel),” Eclipse Arrowhead, IDD 5.2.2, 2026, gateTunnel-v5.2.2-gateTunnelManagement-HTTP-IDD.pdf.

[8] ——, “sysMonitor – Service Description (GateTunnel),” Eclipse Arrowhead, SD 5.2.2, 2026, gateTunnel-v5.2.2-sysMonitor-SD.pdf.

[9] ——, “sysMonitor AMQP/TLS/JSON – Interface Design Description (GateTunnel),” Eclipse Arrowhead, IDD 5.2.2, 2026, gateTunnel-v5.2.2-sysMonitor-AMQP-IDD.pdf.

[10] ——, “ServiceRegistry – System Description,” Eclipse Arrowhead, SysD 5.2.2, 2026, ah5-core-java-spring/serviceregistry/documentation/v5.2.2/ServiceRegistry-v5.2.2-SysD.pdf.

[11] ——, “serviceDiscovery – Service Description (ServiceRegistry),” Eclipse Arrowhead, SD 5.2.2, 2026, ah5-core-java-spring/serviceregistry/documentation/v5.2.2/ServiceRegistry-v5.2.2-serviceDiscovery-SD.pdf.

[12] ——, “Authentication – System Description,” Eclipse Arrowhead, SysD 5.2.2, 2026, ah5-core-java-spring/authentication/documentation/v5.2.2/Authentication-v5.2.2-SysD.pdf.

[13] ——, “identity – Service Description (Authentication),” Eclipse Arrowhead, SD 5.2.2, 2026, ah5-core-java-spring/authentication/documentation/v5.2.2/Authentication-v5.2.2-identity-SD.pdf.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-25	5.2.2	Independent Java build, private helpers and state, per-system deployment and configuration.	Codex (draft)
3	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): standalone deployment, configuration and security statements updated.	Claude Opus 5.5 (draft)
4	2026-11-08	5.2.2	Implementation sections updated for the Java-Spring port (common library, MySQL via JPA, Spring beans for the relay and AH-GT-1); contracts unchanged.	Claude Opus 5.5 (draft)

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	

1 GateTunnel

1.1 System overview

Authorized legacy-mTLS relay and tunnel sessions. The native Go executable selects this system independently; no Java or Python process implements its operations. Shared source is in `internal/core`; federation code is in `internal/relay`.

1.2 Produced and consumed services

Produced HTTPS services: `echo`, `gateTunnelConnection`, `gateTunnelKey`, `gateTunnelManagement`, `generalManagement`. HTTPS service version 5.2.2. Optional native AMQPS telemetry produces `sysMonitor`; SysMonitor additionally publishes `sysMonitorData` over MQTT/TLS. Consumed authorities include ServiceRegistry, Authentication and ConsumerAuthorization; optional dependencies are explicit pinned HTTPS origins.

1.3 Important delimitations

This implementation is a development profile, not a certified interchangeable Java replacement. All systems use private JSON snapshots rather than the Java databases. MQTT application APIs, exhaustive DTO normalization and error parity, automatic data migration and production qualification remain outstanding.

- Legacy-mTLS relay sessions are supported; control-session records are not a second AMQP tunnel data plane. AH-GT-1 encrypted requests are rejected. Consumer listeners are volatile and renegotiated after restart.
- The source reference uses a shared non-Spring HTTP dispatcher. Its route coverage is not inferred from Go route presence; manual JavaSupportRuntime review is required.

1.4 Security and persistence

Mutual TLS verifies the configured cloud trust anchors. Application identity tokens must match the client certificate. Authorities are pinned by system CN/profile before credentials are sent. Management requires a verified sysop identity or an explicit broad management whitelist. The default service authorization policy is enforce; identity-only mode must be explicitly selected. No insecure listener or declared-identity mode is implemented. Each executable owns one locked state file, written through fsync and atomic rename. Cross-system operations are not distributed transactions.

2 Implementation

Go 1.26 or newer builds the native binary. State is serialized as private JSON documents; updates clone state, validate, persist and then publish. Server timeouts and bounded message sizes constrain resource use. Persistent outboxes are at-least-once and cannot provide distributed exactly-once delivery. Detailed source, configuration and test instructions are in the repository README and verification report.

2.1 HTTPS interface operations

The listener is HTTPS with JSON by default; raw configuration and CRL responses use their declared media types. A request body is capped at 1 MiB. The table records implemented routes, not a claim of complete

behavioral conformance.

Method	Path	Access
GET	/gatetunnel/echo	identity
GET	/gatetunnel/publickey	identity
POST	/gatetunnel/connect_provider	identity
POST	/gatetunnel/connect_consumer	identity
POST	/gatetunnel/close_sessions	identity
GET	/gatetunnel/mgmt/sessions	management
POST	/gatetunnel/mgmt/sessions/close	management
POST	/gatetunnel/intercloud/provider	identity
POST	/gatetunnel/intercloud/consumer	identity
GET	/gatetunnel/mgmt/status	management
GET	/gatetunnel/monitor/echo	certificate
GET	/gatetunnel/general/mgmt/get-config	management
POST	/gatetunnel/general/mgmt/logs	management

1 GateTunnel

1.1 System overview

Native Rust development profile for local Java source `JavaSupportRuntime/`. One independently configured executable, identity, port and state file per role. No Java, Go or Python process implements the service. Some essential operations are explicitly unavailable.

1.2 Design and trust boundaries

Tokio/Hyper/Axum with rustls enforces mutual TLS. Certificate chain, validity, local CN/profile and startup-configured optional CRLs are checked. CRLs need process restart to refresh; without CRLs no revocation enforcement is claimed. Tokens are verified with Authentication and bound to the TLS identity. Business calls require ConsumerAuthorization; management requires a verified sysop or explicit broad whitelist. CA provisioning instead applies Java-style named certificate-principal rights. Outbound authorities are pinned to expected CN/profile before credentials are sent, in addition to chain, SAN and time validation.

1.3 Persistent state

Independent locked atomic JSON replaces SQL. Clone/validate/fsync/rename commits prevent partially published local transactions. Rename is the commit point; directory-sync failure is an uncertain durable commit, not a rollback. Owner-only keys/password files are separate. Network effects are not distributed transactions; interrupted starts and bridge operations require reconciliation. No automatic Java state import, clustering or deployment is delivered.

1.4 Implemented behavior

Public key and empty session catalogue

1.5 Open release gates

Session allocation/listeners/relay data plane absent, returns501. All systems remain development profiles, not certified interchangeable Java replacements. See `COMPATIBILITY.md` for detailed DTO fields and restrictions; the Java sources remain the authoritative contract.

1.6 Produced and consumed services

Advertised implemented HTTPS definitions: `echo`, `gateTunnelKey`, `gateTunnelManagement`, `generalManagement`.

Authorities consumed as applicable: Authentication, ServiceRegistry, ConsumerAuthorization and explicitly configured pinned adapter/provider origins. Only implemented operations are registered. Service versions are 1.0.0 except Gatekeeper/GateTunnel 5.2.2. No AMQP sysMonitor.observe implementation is advertised.

1.7 HTTPS operations

JSON unless raw configuration bytes, PEM CRL or echo text. Bodies: 1 MiB; remote responses: 8 MiB. Header: `Bearer IDENTITY-TOKEN//token`. Unknown routes return 404, absent dependencies 503. Unported

operations return 501 after applicable access checks and are excluded from registration. Route presence is not semantic parity.

Method	Path	Access / status
GET	/gatetunnel/general/mgmt/get-config	management
POST	/gatetunnel/general/mgmt/logs	management
GET	/gatetunnel/monitor/echo	certificate
GET	/gatetunnel/publickey	identity
POST	/gatetunnel/connect_provider	identity / 501
POST	/gatetunnel/connect_consumer	identity / 501
POST	/gatetunnel/close_sessions	identity / 501
GET	/gatetunnel/mgmt/sessions	management
POST	/gatetunnel/mgmt/sessions/close	management / 501

Error envelope: errorMessage, errorCode, exceptionType, origin and timestamp. Full Java status, DTO and validation equivalence requires differential testing.

2 References and qualification

See `contracts.json` for local Java SHA256 hashes/mappings, `COMPATIBILITY.md` for DTO details and gaps, and `VERIFICATION.md` for tests actually run. No upstream release-tag provenance, complete replacement or real-cloud interoperability is asserted.